Smart & Green Gaming: Using GenAI to Automate Mobile Game Energy Testing

Cristiano Zhou[1], Wellington Oliveira[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

reliable software systems

CONTEXT

How Android bills energy

1. Power profile estimation
   → no coefficient per component
   → components may not be billed
2. The screen is billed by its brightness

OBJECTIVE

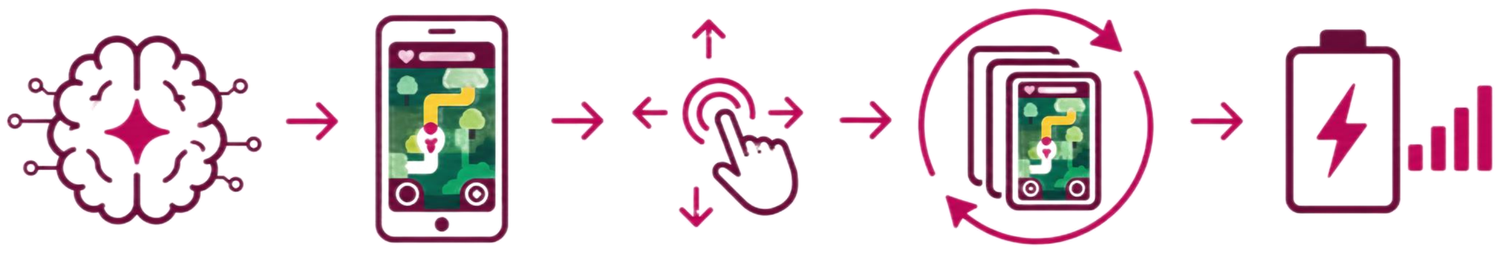
- **Games → most power-hungry apps**
- **Manual play → slow, not repeatable**
- **Fixed scripts → break on pop-ups**

THE GOAL

Use generative AI to automate these tests: from screenshots, an LLM writes the code → the game plays itself, repeatedly, through ebserver

Why games break UI automation

- One drawing surface → no UI elements
- Fixed coordinates → break on pop-ups



PIPELINE

PHASE 1 · LLM writes the script

Screenshot via ADB → **LLM** next ADB command → **Python script** emulator prototype → repeat until stopping point

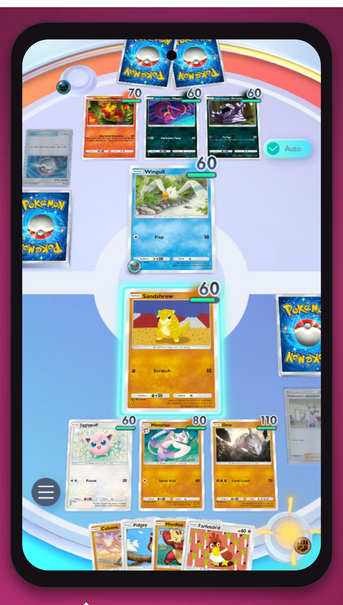
PHASE 2 · Repeated measurement

Java test UI Automator → **ebserver** 30 runs, real phone → **batterystats** mAh per run → **CSV + plots** extract_csv.py

GAMES TESTED



Flow Free
animations on / off



Pokémon TCG Pocket
quality × fps



Sky Arena
speed × reduce mode

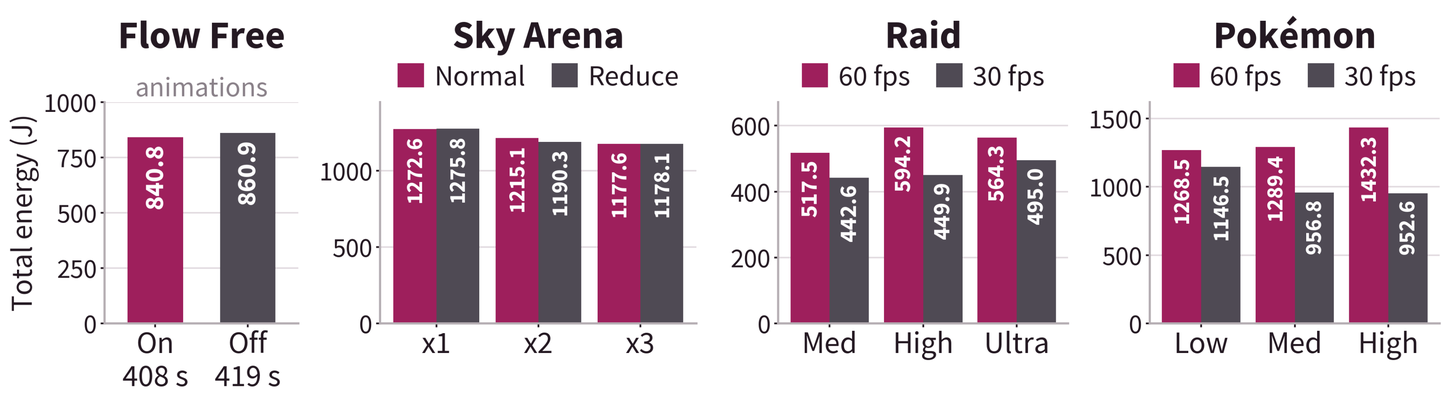


Raid: Shadow Legends
quality × fps

RESULTS

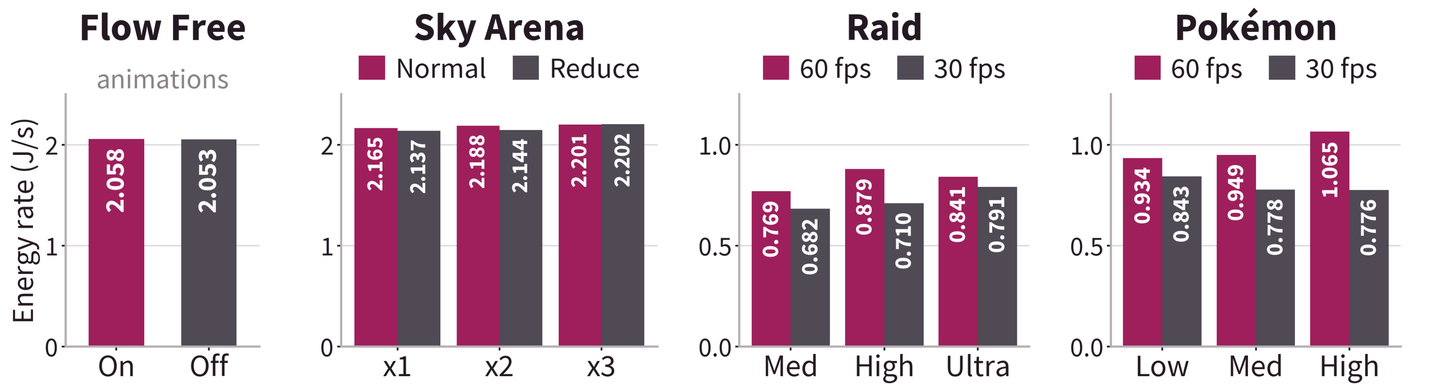
20 configurations × 30 runs - Samsung SM-G780G

TOTAL ENERGY (J)



Flow Free “off” runs 11 s longer (+2.4 %). Sky Arena x3 finishes sooner (−7.5 %)

ENERGY PER SECOND (J/s)



Animations, speed and reduce mode barely move the rate; 30 fps lowers it every time

CONCLUSIONS

AUTOMATION

LLM-built tests run 30 times fully automated

SETTINGS

60 → 30 fps saves 6–27 %
Animations/reduce mode: no effect

LIMITS

- Variance between runs
- Single device

NEXT STEPS

More games, more devices, smarter exploration

Acknowledgements: This work was supported by the LASIGE Research Unit, ref. UID/000408/2025.